

In []:

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In [1]: %matplotlib ipynb
import numpy as np
import matplotlib.pyplot as plt
```

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In [6]: # define the differential equation
def f(r, t): # don't forget r is a vector!
    x = r[0]
    v = r[1]
    g = 9.8 # N/kg
    m = 1 # kg
    c = 0.05 # what are these units?
    fx = v # this is the dx/dt equation!
    fv = -g - c/m*(v**2) # this is the dv/dt equation!
    return(np.array([fx, fv]))

# define the boundary condition
a = 0.0
b = 1.414
N = 100
dt = (b-a)/N

# Runge-Kutta 4th order

r = np.array([10.0, 0.0]) # initial condition

tpoints = np.arange(a, b, dt)

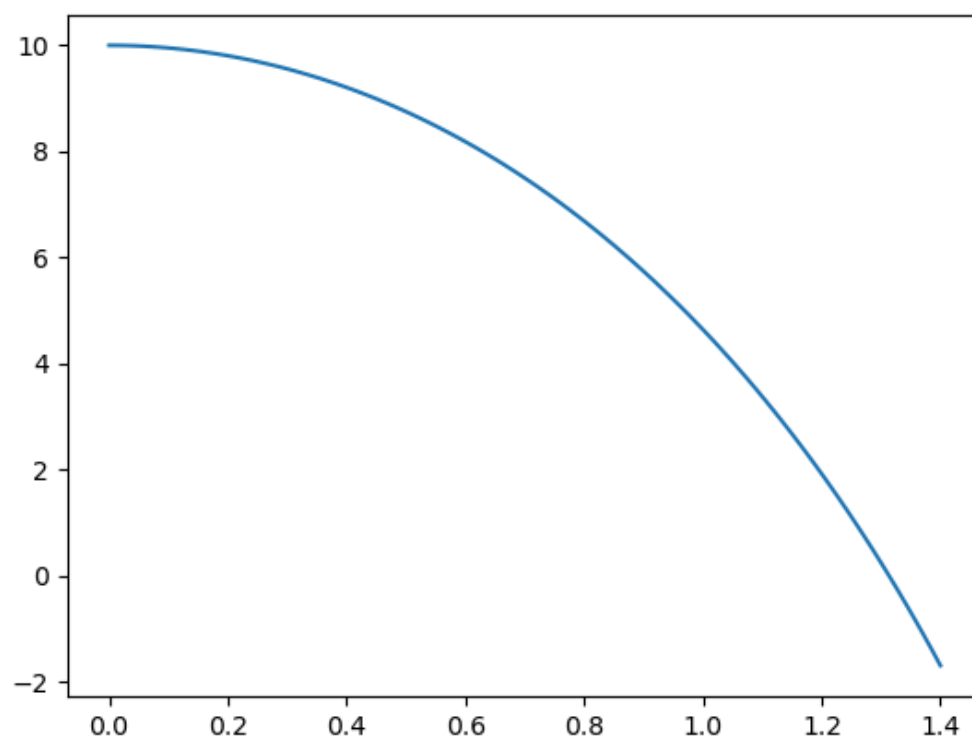
xpointsD2 = []
vpointsD2 = []

for i in tpoints:
    xpointsD2.append(r[0])
    vpointsD2.append(r[1])
    k1 = dt*f(r, i)
    k2 = dt*f(r+1/2*k1, i+1/2*dt)
    k3 = dt*f(r+1/2*k2, i+1/2*dt)
    k4 = dt*f(r+k3, i+dt)
    r = r + (k1+2*k2+2*k3+k4)/6

fig0, ax0 = plt.subplots()
ax0.plot(tpoints, xpointsD2)
#ax0.plot(tpoints, vpointsD)
#ax0.plot(tpoints, vpoints)
```

Out[6]: [<matplotlib.lines.Line2D at 0x7951c5bb8770>]

Figure



In []: